ModelIt!

Student Activity Pack

Designing Organisms That Eat Plastic

Synthetic Biology and Metabolic Engineering for Environmental Remediation

Grade 09L3 | NGSS HS-LS1-5, HS-ETS1-2 | Energy and Matter / Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Do you think any living organism can naturally break down plastic — and if not, why not?

2 What would you need to change about a bacterium to make it capable of eating plastic?

3 Draw a diagram showing how you imagine an engineered plastic-eating organism would work at the molecular level.

4 What risks might come from releasing genetically engineered organisms into the ocean?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Enzyme Activity Level
- Substrate Concentration
- Metabolic Pathway Efficiency
- Byproduct Toxicity
- Organism Growth Rate
- Environmental Temperature
- pH Tolerance
- Oxygen Requirements
- Degradation Rate
- Containment Level

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Controlled Lab

Ideal conditions, maximum containment

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Ocean Deployment

Real ocean conditions, reduced containment

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Optimized Field Release

Balanced performance and safety

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Biofilm Formation Capacity — The engineered organism's ability to form biofilms on plastic surfaces, which concentrates enzyme activity at the substrate interface and dramatically improves degradation efficiency compared to free-floating organisms

Horizontal Gene Transfer Risk — The probability that the engineered genes transfer from the designed organism to wild microbes through natural DNA exchange mechanisms — a containment breach that could create unintended environmental consequences

Multi-Plastic Capability — The breadth of plastic types the engineered organism can degrade — PET, polyethylene, polypropylene, polystyrene, and nylon all require different enzymes, and a single organism that handles multiple types would be far more practically useful

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

8 million tons of plastic enter the ocean every year. Can we engineer a living organism that breaks it down? Use evidence from your simulation to support your answers.

1. **How does your model reveal the gap between laboratory enzyme performance and real-world environmental degradation?**

2. **What is the containment-effectiveness paradox, and how does your model help navigate it?**

3. **Why is Metabolic Pathway Efficiency critical for preventing the cure from being worse than the disease?**

4. **How would adding Biofilm Formation Capacity change the organism's environmental performance?**

5. **What ethical responsibilities do synthetic biologists have when designing organisms intended for environmental release?**

8 STEM Challenge: Design a Plastic-Eating Organism for Ocean Cleanup

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YOUR ENGINEERING MISSION

Design an engineered biological system that degrades ocean plastic waste at meaningful rates while maintaining rigorous biosafety containment.

THE SCENARIO

An environmental biotech company has received a $100 million grant to develop a biological plastic cleanup system for the Great Pacific Garbage Patch. Current mechanical cleanup methods remove only a fraction of the 80,000 tons of plastic accumulated there. Your synthetic biology team must design an organism-based system that can degrade plastic at scale in open ocean conditions while meeting regulatory biosafety requirements.

Engineering Design Thinking

1. WOULD YOU DEPLOY FREE-LIVING ORGANISMS OR ATTACH THEM TO FLOATING DEGRADATION PLATFORMS?
2. HOW WOULD YOU HANDLE THE TOXIC BYPRODUCTS GENERATED DURING PLASTIC BREAKDOWN?
3. WHAT GENETIC KILL SWITCHES OR AUXOTROPHIES WOULD YOU ENGINEER FOR CONTAINMENT?
4. HOW WOULD YOU MONITOR ORGANISM SPREAD AND DEGRADATION PROGRESS IN THE OPEN OCEAN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Synthetic Biologists design and engineer organisms with novel capabilities for applications in medicine, energy, materials, and environmental remediation. They work for biotech companies (Ginkgo Bioworks, Zymergen), environmental engineering firms, and academic research labs, earning $85,000–$170,000/year. Bioremediation Engineers who deploy biological cleanup systems earn $75,000–$140,000/year.